

NICE on VTE

1. Diagnosis of DVT

NICE NG158, 2023

Can be used in those with COVID-19 and suspected VTE



Red Whale

GEMS
Guidelines & Evidence Made Simple

The whole NICE guideline applies to those $\geq 18y$ but NOT to pregnant women.

DVT: diagnosis

ADAPT THIS TO YOUR LOCAL PATHWAY and access to D-dimer. Start by doing a Wells' DVT score and if:

- DVT unlikely on Wells' score: do D dimer; if positive, do a scan.
- DVT likely on Wells' score: do scan; if negative, do D-dimer.

If initial test (scan/D-dimer) is to be delayed by more than 4h, anticoagulate in the interim (take D-dimer first!).

DVT suspected: **do Wells' DVT score (see below)**

Wells' DVT score ≤ 1

Do D-dimer. Consider age-adjusting result: see below. If result not available within 4h:

- Offer interim anticoagulation

D-DIMER POSITIVE

Do proximal leg vein ultrasound.

If not available within 4h, should scan within 24h AND offer interim coagulation

SCAN NEGATIVE

SCAN POSITIVE

DVT UNLIKELY

- Stop interim anticoagulation
- Consider alternative diagnoses

D-DIMER NEGATIVE

Wells' DVT score ≥ 2

Do proximal leg vein ultrasound.

If not available within 4h, should scan within 24h AND:

- Do D-dimer
- Offer interim coagulation

SCAN NEGATIVE

Do D-dimer. Consider age-adjusting result: see below

D-DIMER POSITIVE

Stop interim anticoagulation.

Repeat scan 6-8d later

Repeat scan
NEGATIVE

Repeat scan
POSITIVE

DVT CONFIRMED: ANTICOAGULATE

(if contraindicated, offer mechanical interventions, e.g. inferior vena caval filters, percutaneous thrombectomy)

If iliofemoral DVT: thrombolytic therapy may be considered

SCAN POSITIVE

D-dimer can be kept overnight in the fridge

Before starting anticoagulation, take bloods (but don't wait for results!):
FBC, renal, LFTs, PT, APTT

Age-adjusting D-dimer tests in VTE: NICE says "consider adjusting D-dimer" if $>50y$. Common formulas are:

If you use FEU units (your upper limit of normal is around 400/450/500mcg/l): age in years $\times 10$ (= 600 if 60y).

If you use DDU units (your upper limit of normal is around 200/230/250mcg/l): age in years $\times 5$.

DVT Wells' score (not to be confused with Wells' PE score!)	Score	Interpretation
Cancer (on treatment or treatment $<6m$ ago or palliative)	+1	Score ≥ 2 : DVT likely Score ≤ 1 : DVT unlikely
Previous DVT/PE	+1	
Bed-bound for more than 3d OR major surgery in last 12w	+1	
Immobility (plaster), paralysis, paresis of leg	+1	
Localised tenderness along deep venous system	+1	
Entire leg swollen	+1	
Calf circumference $>3cm$ than other leg	+1	
Pitting oedema confined to symptomatic leg	+1	
Collateral superficial veins (not longstanding varicose veins)	+1	
Alternative diagnosis as likely or more likely than DVT*	-2	

*What does this mean? Imagine I have a nasty cut on my calf which starts to look infected and there is significant redness and swelling around it. The most likely diagnosis is cellulitis, even though there are some features of a DVT, so I score -2.

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NICE on VTE

2. Diagnosis of PE

NICE NG158, 2023

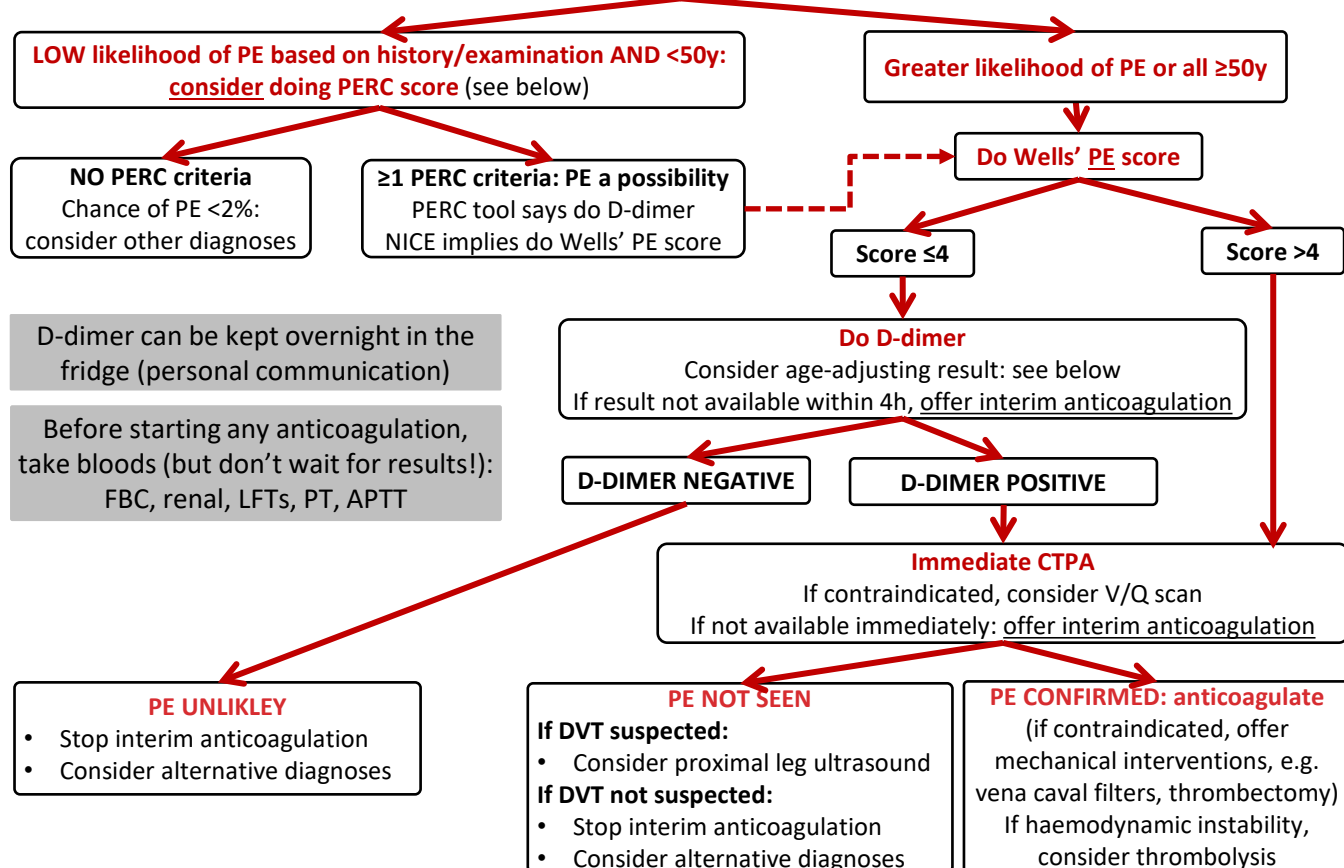
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PE: diagnosis PE suspected (e.g. chest pain, shortness of breath, haemoptysis). Examine. Consider CXR to exclude other causes (at Red Whale, we think we should not wait for CXR result if result not available fairly immediately). If clinical suspicion of PE is low (from history, examination and any investigations (such as ECG/CXR)) and other diagnoses are possible, consider using PERC to decide if further investigations needed (skip this step if $\geq 50y$ as PERC will be positive). **LOCAL PATHWAYS VARY: some can do point-of-care D-dimers, others can't. ADAPT THIS TO YOUR LOCAL PATHWAY.**



Age-adjusting D-dimer tests in VTE: NICE says "consider adjusting D-dimer" if $>50y$. Common formulas are:
If you use FEU units (your upper limit of normal is around 400/450/500mcg/l): age in years $\times 10$ (= 600 if 60y).
If you use DDU units (your upper limit of normal is around 200/230/250mcg/l): age in years $\times 5$.

PERC (PE rule-out criteria)	Interpretation	
Age $\geq 50y$	If NONE of the features listed, score 0: chance of PE $<2\%$; consider other diagnoses. If ANY of these features present, continue to investigate for PE. <i>N.B. has not been tested in those with COVID-19.</i> NICE recommended using PERC to reduce anxiety, D-dimer testing, imaging and interim anticoagulation. Evidence base limited, and all from A&E trials, but NICE thinks can be used in other settings.	
HR ≥ 100		
Oxygen sats on air $<95\%$		
Unilateral leg swelling		
Haemoptysis		
Recent surgery or trauma ($\leq 4w$ ago, requiring GA)		
Exogenous hormone use (contraceptives, HRT)		
PMH DVT/PE		
PE Wells' score (not to be confused with Wells' DVT score!)	Score	Interpretation
Clinical signs/symptoms of DVT (minimum of leg swelling and pain on palpation of deep veins)	3	Score >4 : PE likely Score ≤ 4 : PE unlikely
Alternative diagnosis LESS likely than PE	3	
P >100	1.5	
Immobilisation $>3d$ or surgery in last 4w	1.5	
Previous DVT/PE	1.5	
Haemoptysis	1	
Cancer (on treatment or treatment $<6m$ ago or palliative)	1	

3. Treatment of DVT and PE (NICE)

(applies to those 18y but NOT to pregnant women)

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Anticoagulation for DVT/PE

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- Before starting anticoagulation, take the following bloods (but don't wait for the results!): FBC, renal function, LFTs, PT and APTT. Review and act on blood test results within 24h of starting anticoagulation.

INTERIM anticoagulation if suspected DVT/PE (i.e. where there is a delay waiting for tests to establish the diagnosis)

- Choose (if possible) an anticoagulant that can be continued if the diagnosis is confirmed (likely to be a DOAC, unless good reason not to, e.g. significant renal impairment).
- When interim anticoagulation stopped, do not stop any anticoagulation given for another reason (e.g. those meeting criteria for prophylaxis of VTE in acute COVID).

ANTICOAGULATION FOR CONFIRMED DVT/PE

- Use apixaban or rivaroxaban.** If neither suitable, dabigatran/edoxaban/warfarin can be used. If using dabigatran/edoxaban/warfarin, must load with LMWH (for at least 5d with dabigatran/edoxaban or until warfarin level therapeutic). If using warfarin, do not routinely offer self-monitoring.
- Some PEs are low-risk PE (assessed using scoring tool): outpatient treatment may be an option (specialist decision!).
- When starting anticoagulation, ensure patients are fully informed about:**
 - How to take it and possible side-effects.
 - Interactions with other medication and the need to carry an anticoagulation card at all times.
 - Impact on dental treatment, sport, travel and pregnancy.
- For those concerned about using animal products for religious/ethical reasons or because of food intolerances:**
 - Apixaban and rivaroxaban contain lactose from cows' milk; heparins are of animal origin.

Special situations: specialist team will advise on the best anticoagulant and whether any monitoring is required:

- Extremes of body weight: <50kg or >120kg.
- Renal impairment (note: based on creatinine clearance, calculated using Cockcroft-Gault not eGFR):
 - Creatinine clearance 15–50: may be able to use standard treatments, sometimes with dose adjustment and in line with SPC (DOACs have different thresholds of renal impairment at which they can be used).
 - Creatinine clearance <15: warfarin is usually the treatment of choice, with heparin cover while loading.
- Active cancer: DOAC first choice.
- Triple positive antiphospholipid syndrome (positive for lupus anticoagulant, anticardiolipin and anti-beta2-glycoprotein): DO NOT USE DOACs (seem to be less effective): use WARFARIN, with LMWH cover while loading.

Elasticated graduated compression stockings

- Do NOT use to prevent VTE recurrence or post-thrombotic syndrome. Can be used for post-DVT leg symptoms.

Duration of therapy: MINIMUM 3 months for everyone, but then specialist decision when to stop, usually:

Active cancer (not yet treated, ongoing treatment, not been cured): **treat for 3–6m.** Specialist to decide if to continue beyond this. (non-active cancer: treat as for everyone else)

Provoked VTE:
3 months treatment if provoking factor is no longer present.
Provoked = transient major risk factor in past 3m.
NICE gives these examples; more info in main article:

- Surgery/trauma.
- Significant immobility (bed bound, can't walk unaided, spending substantial time in bed/chair).
- Hormones: pregnancy/puerperium, combined pill, HRT.

Unprovoked VTE:
treat for 3m then specialist to determine long-term treatment (may be lifelong).
Review this decision annually. If long-term anticoagulation declined, consider aspirin 75mg or 150mg daily.

Investigations for underlying causes (cancer, thrombophilia) after UNPROVOKED VTE

CANCER	Offer physical examination and review baseline bloods (FBC, renal function, LFTs, PT, APPT). Do NOT investigate for cancer (unless they have symptoms!). Rationale explained in main article.
THROMBOPHILIA	- Consider testing those with <u>unprovoked VTE</u> who have a <u>first-degree relative with VTE</u> if the plan is to stop anticoagulation (but tests usually need to be done OFF anticoagulation!). - Do NOT offer testing if: - Provoked VTE. - Continuing long-term anticoagulation (it won't change your management!). - Do NOT test if family history of thrombophilia and VTE BUT they themselves have not had a VTE.

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